FCE_Science_Workshop_3.qmd

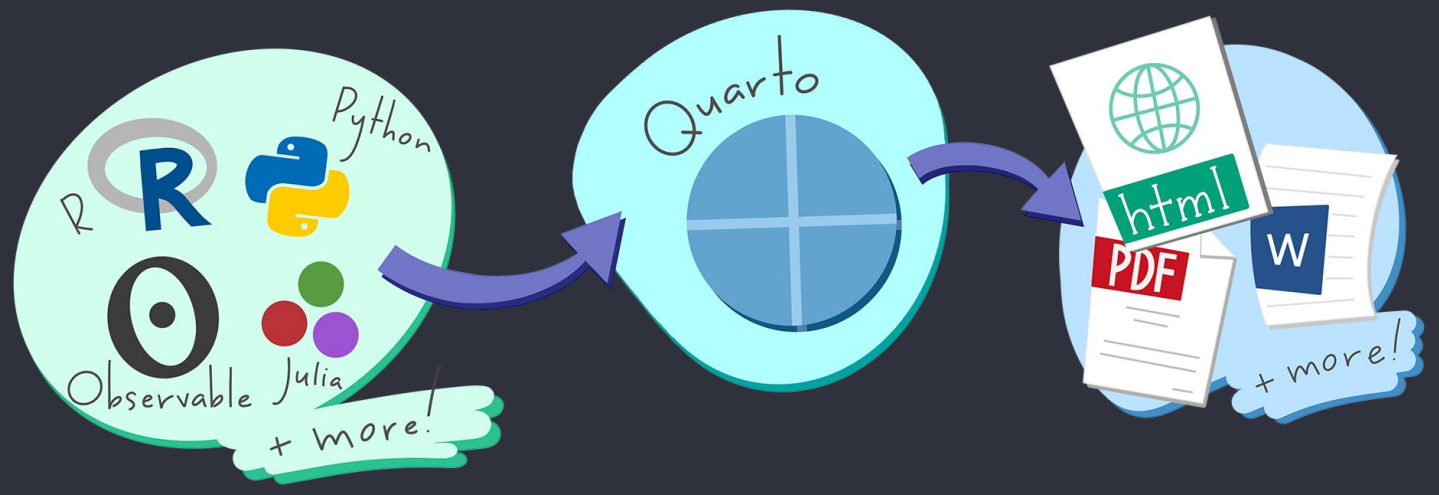
```
# Github Website Making Workshop using Quarto
# Led by: Hannah-Marie Lamle


# March 19th, 2026 3:00pm - 5:00pm
```

- **Introduction to Quarto**
  - What is it?
  - Download Quarto
- **Set Up**
  - Create a new Quarto website
  - Create a GitHub repository
  - Pushing website to Github + launch
- **Design!!**
  - Quarto website files and vocabulary
  - Markdown formatting
  - BONUS: Publishing a Dashboard!!

What is Quarto?

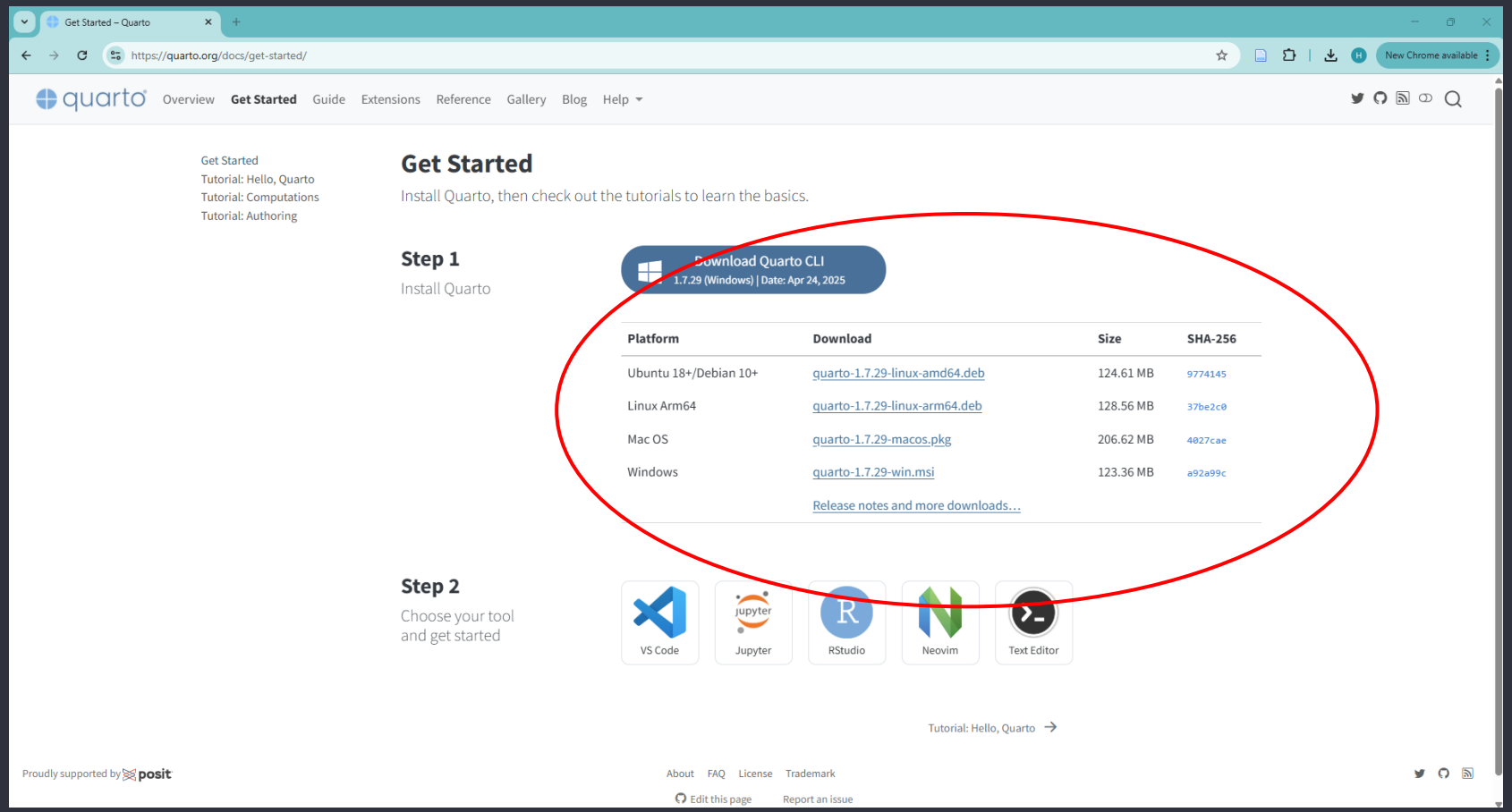
- “An open-source scientific and technical publishing system” - https://quarto.org/
- Similar to LaTex, create documents (markdown files) presentations, blogs, dashboards, websites using a coding tool (python, Rstudio, etc)



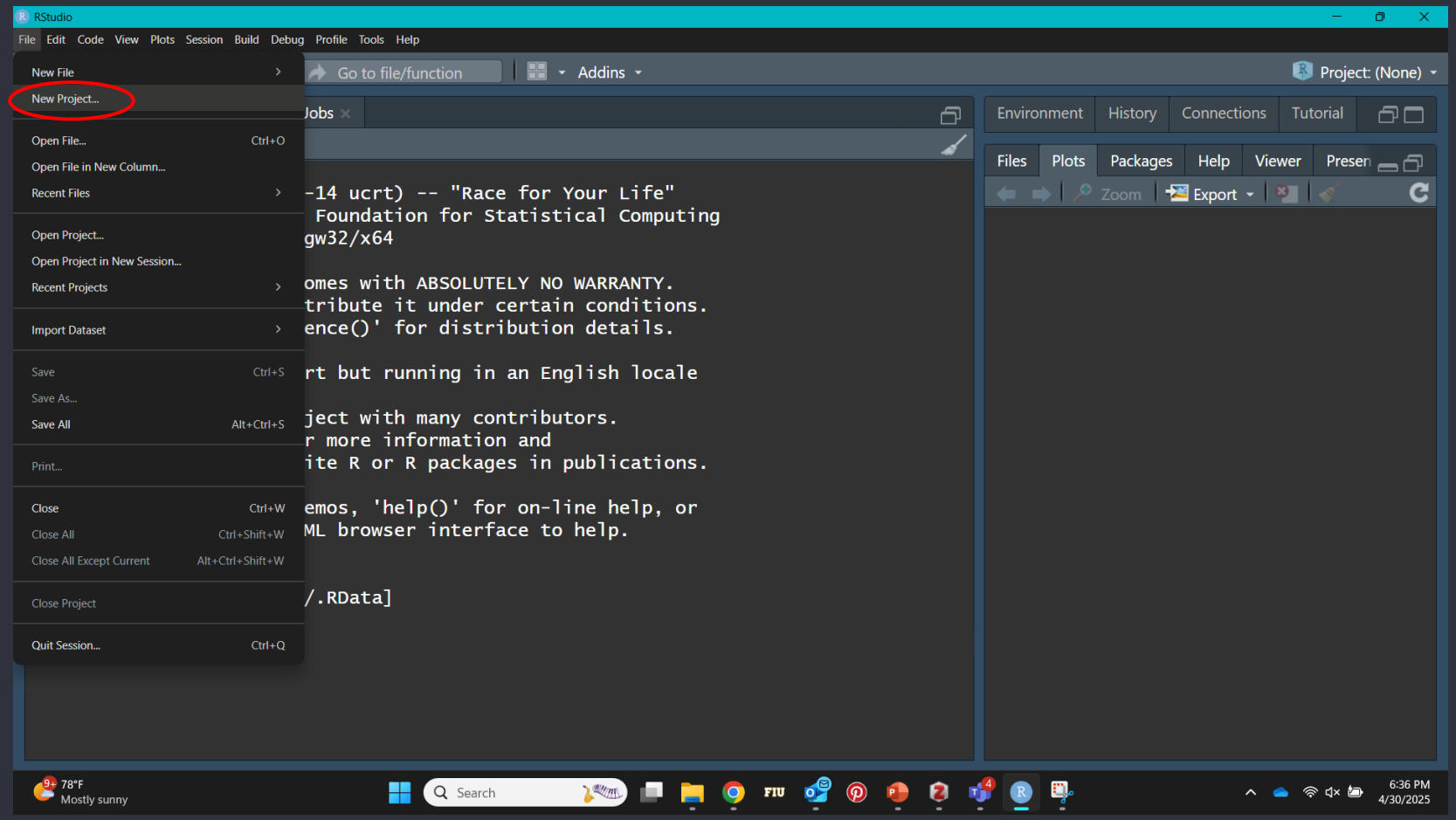
Ready? Let's go!

- LOG OUT of Rstudio if you have it up currently
- Log into Github

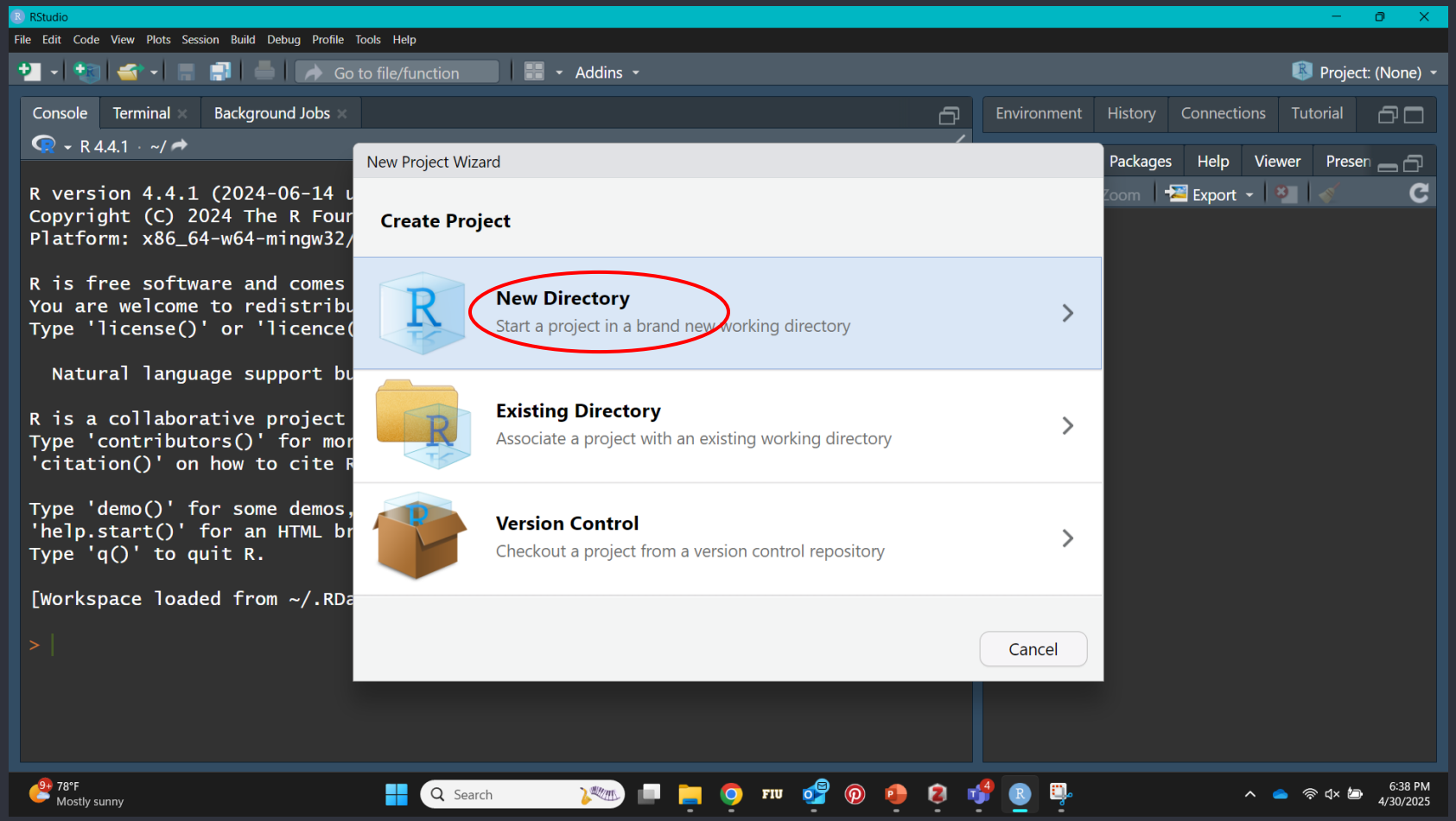
Download Quarto: https://quarto.org/docs/get-started/



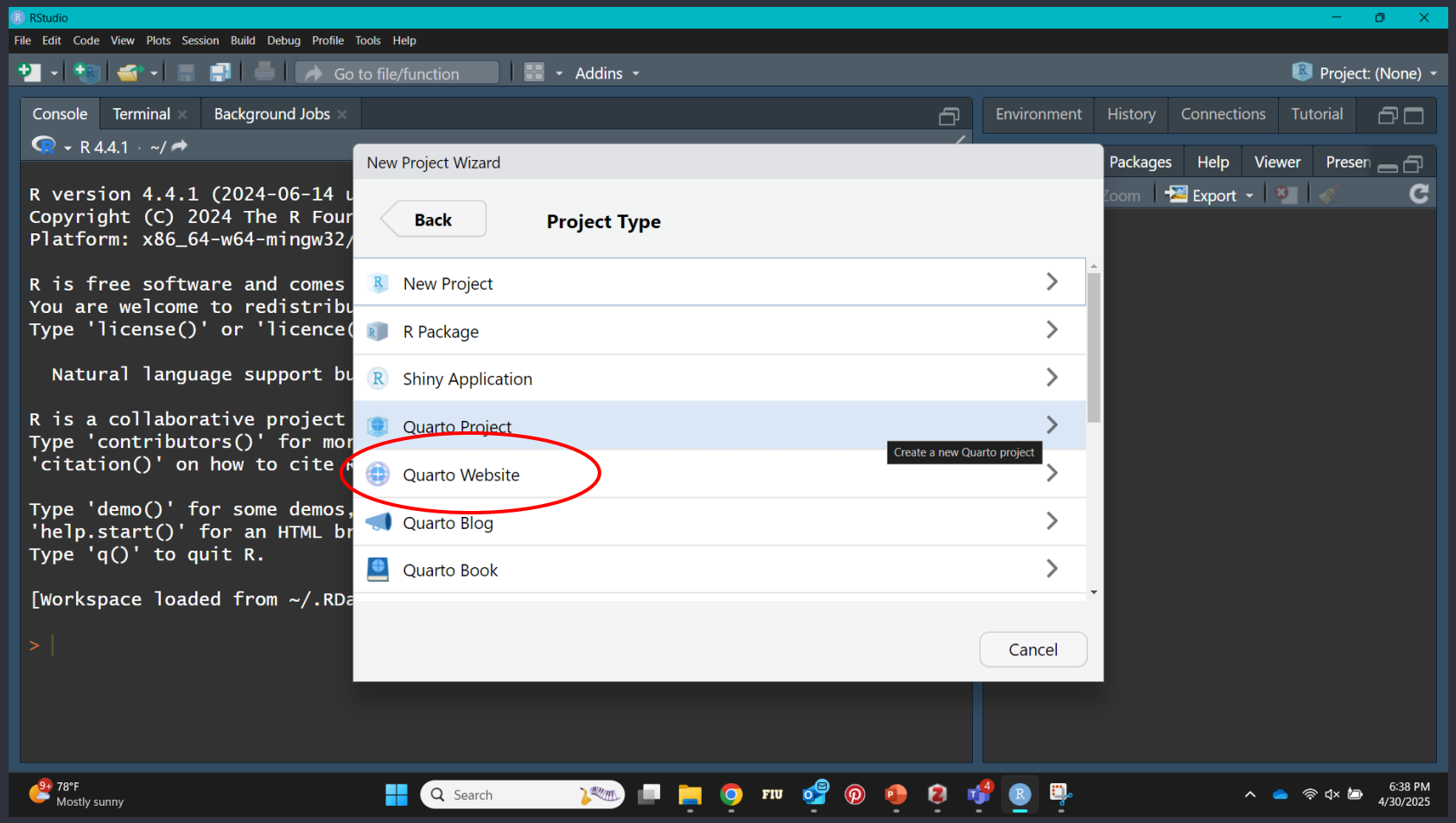
Create a new Quarto Website: file > new project



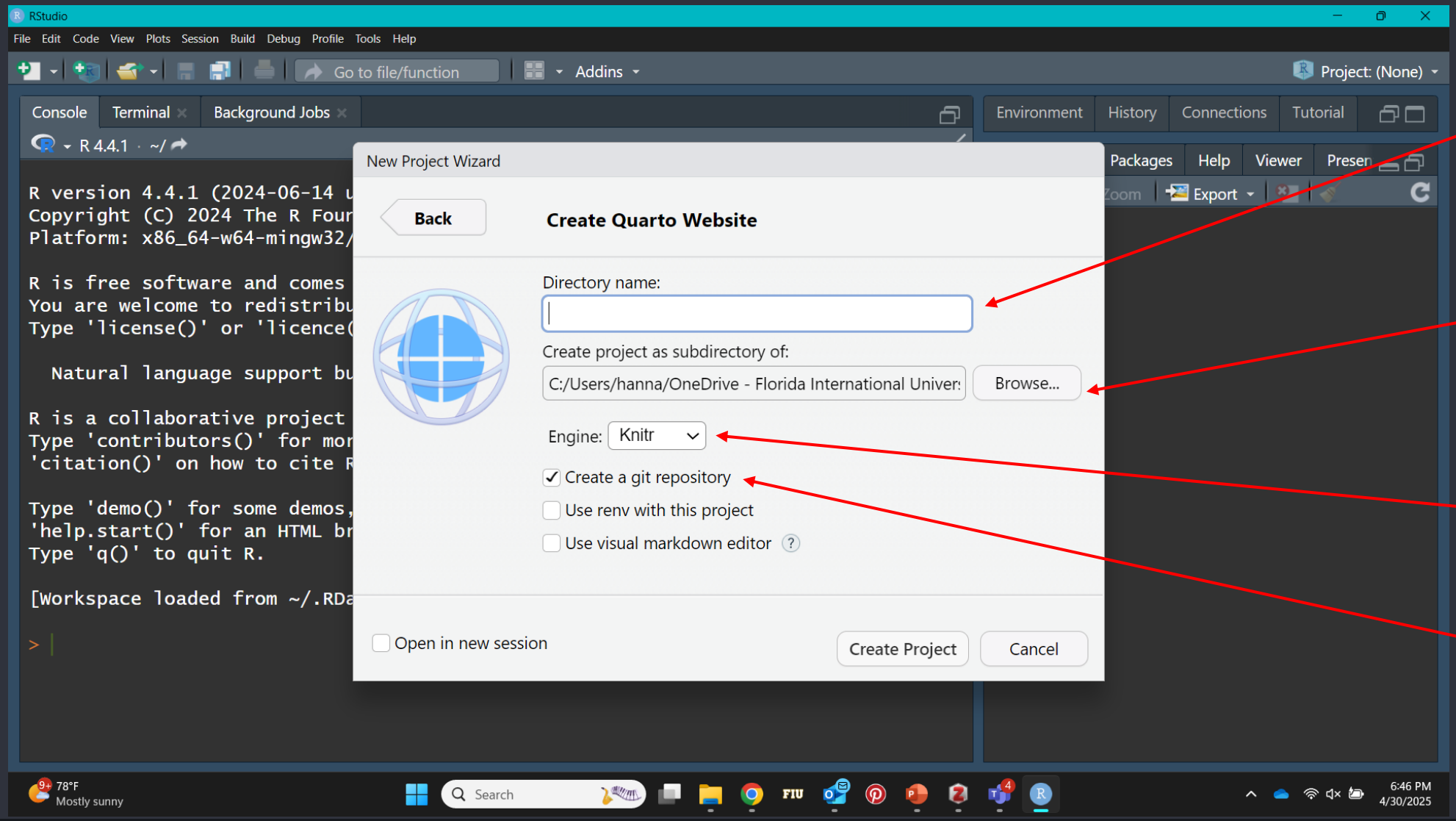
Create a new Quarto Website: New Directory



Create a new Quarto Website: Quarto Website



Set_Up.qmd



Directory name: does NOT have to match GitHub repo "my_website"

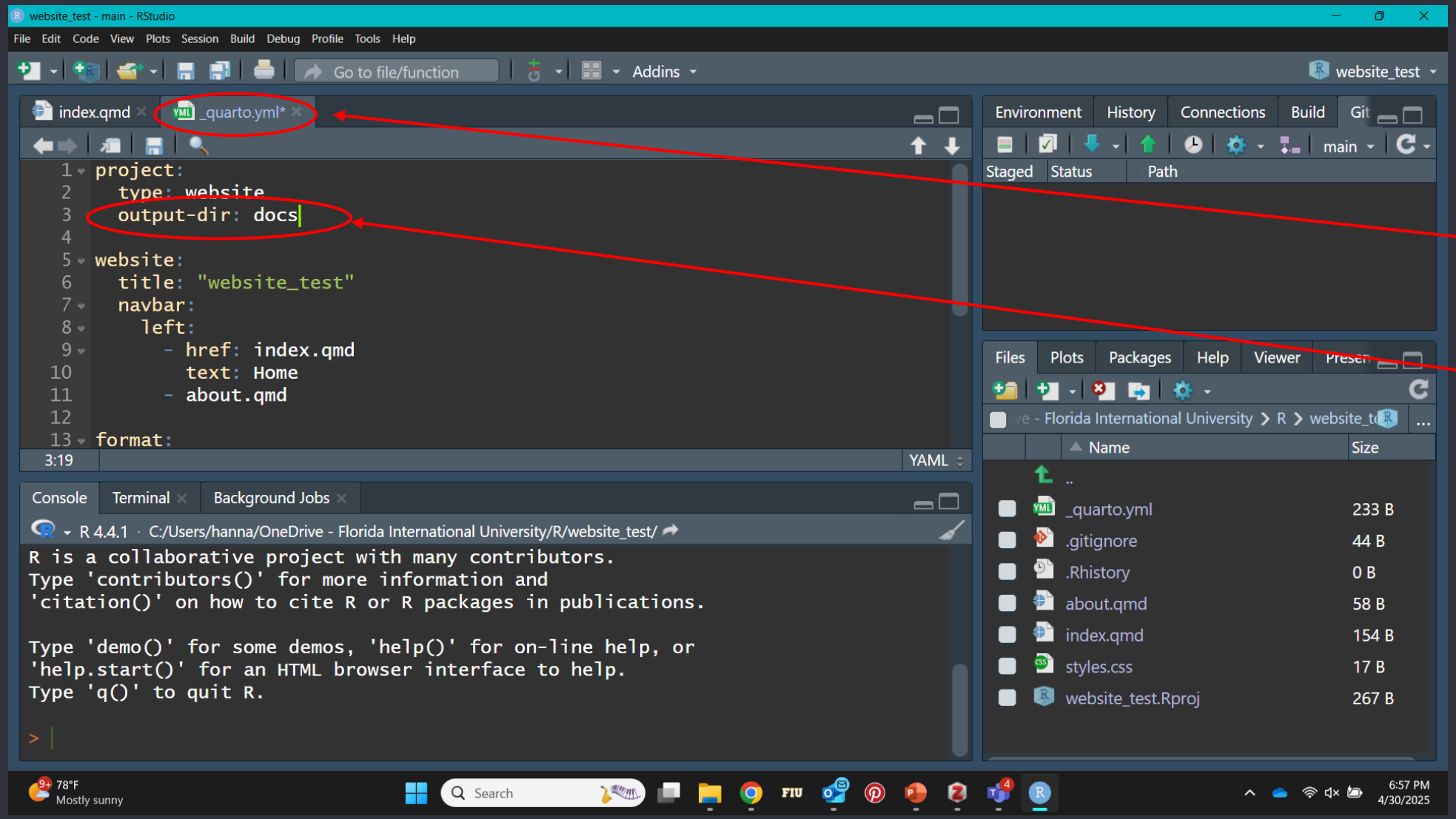
Location for your directory (choose where makes most sense for you)

Should be "Knitr" for everyone

YES!! Makes set up easier

FCE_Science_Workshop_3.qmd

Set_Up.qmd



You have a website!!

1. Go to "_quarto.yml" document

2. Include this line of code. (Indents are important!!)

3. SAVE!!

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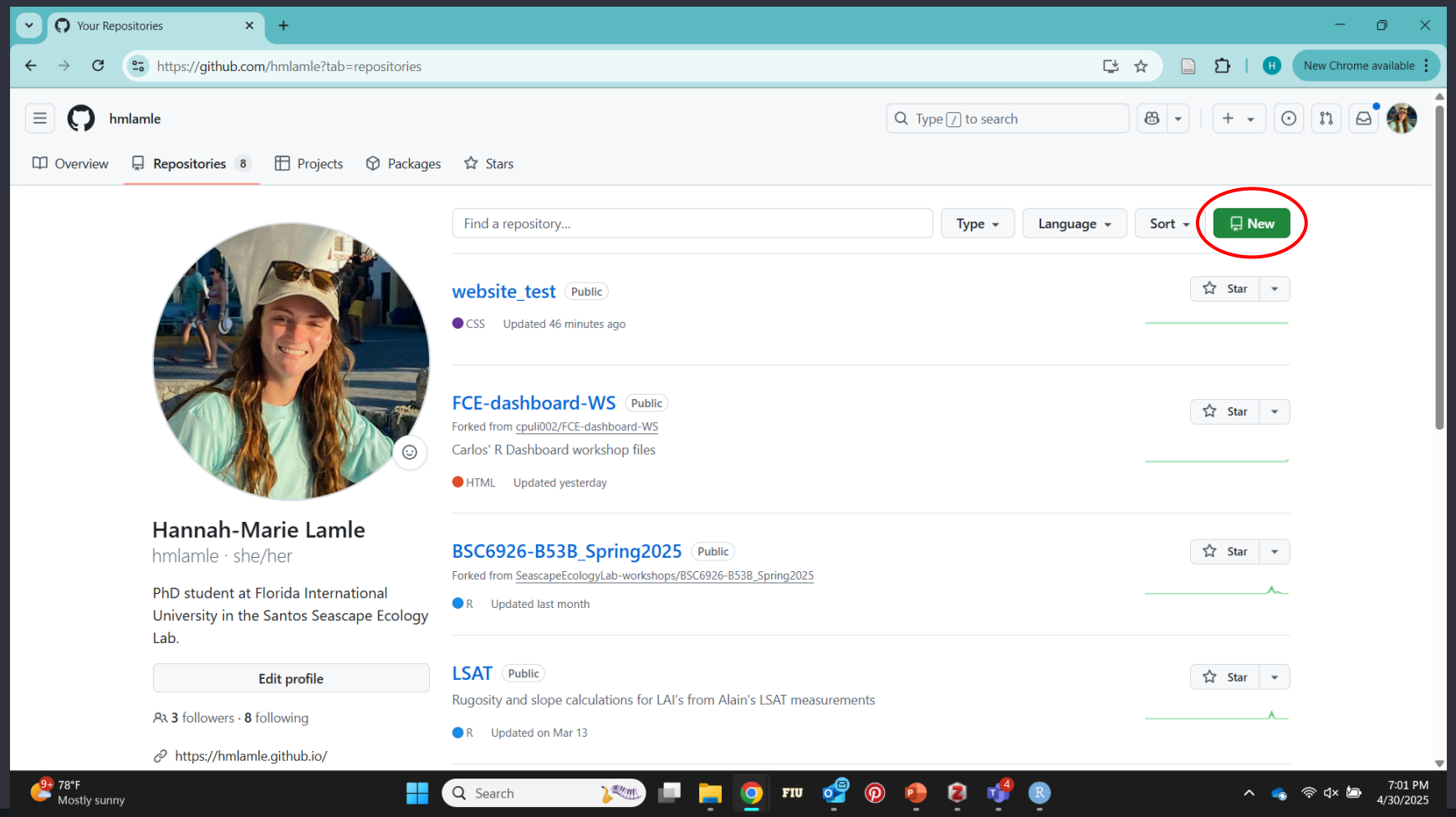
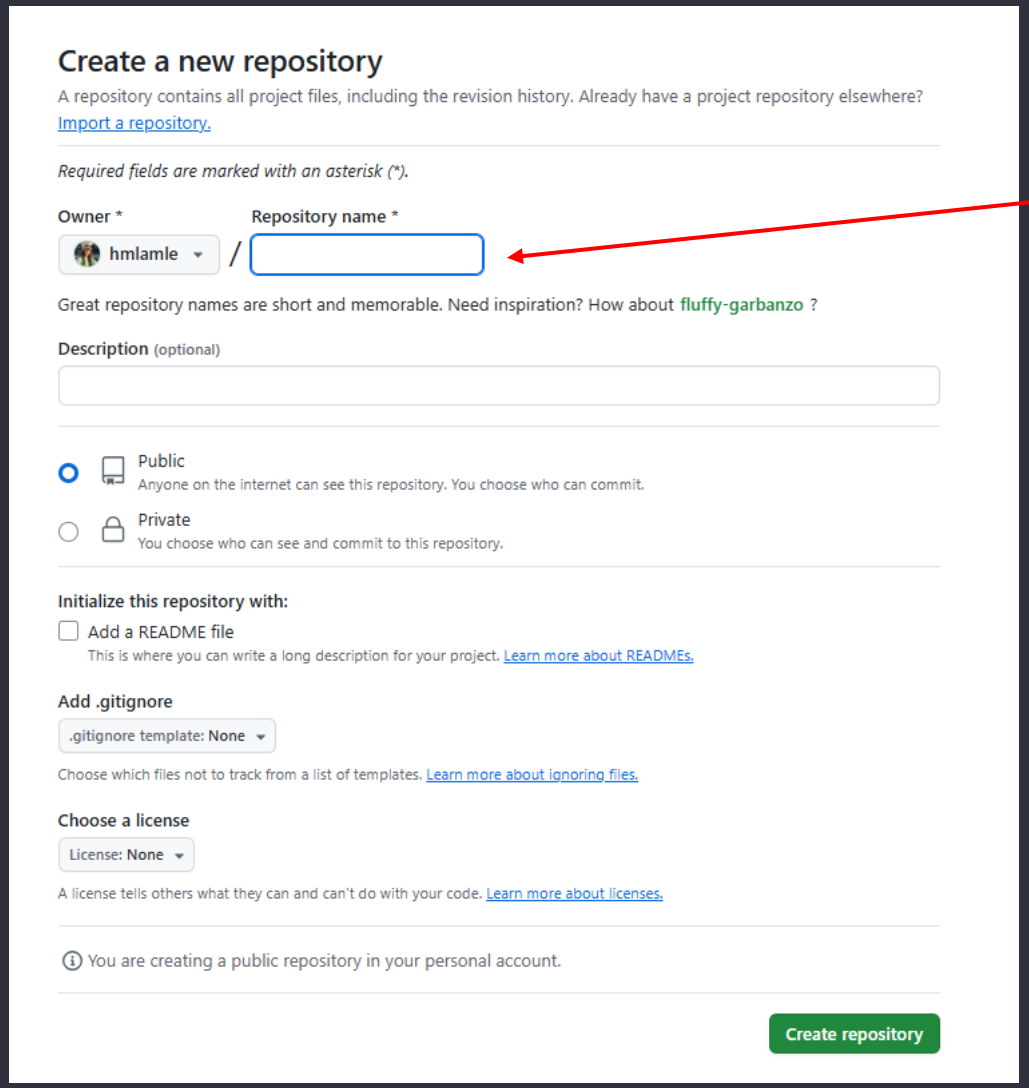
Switch to GitHub.. Create new repository to host website



Set_Up.qmd



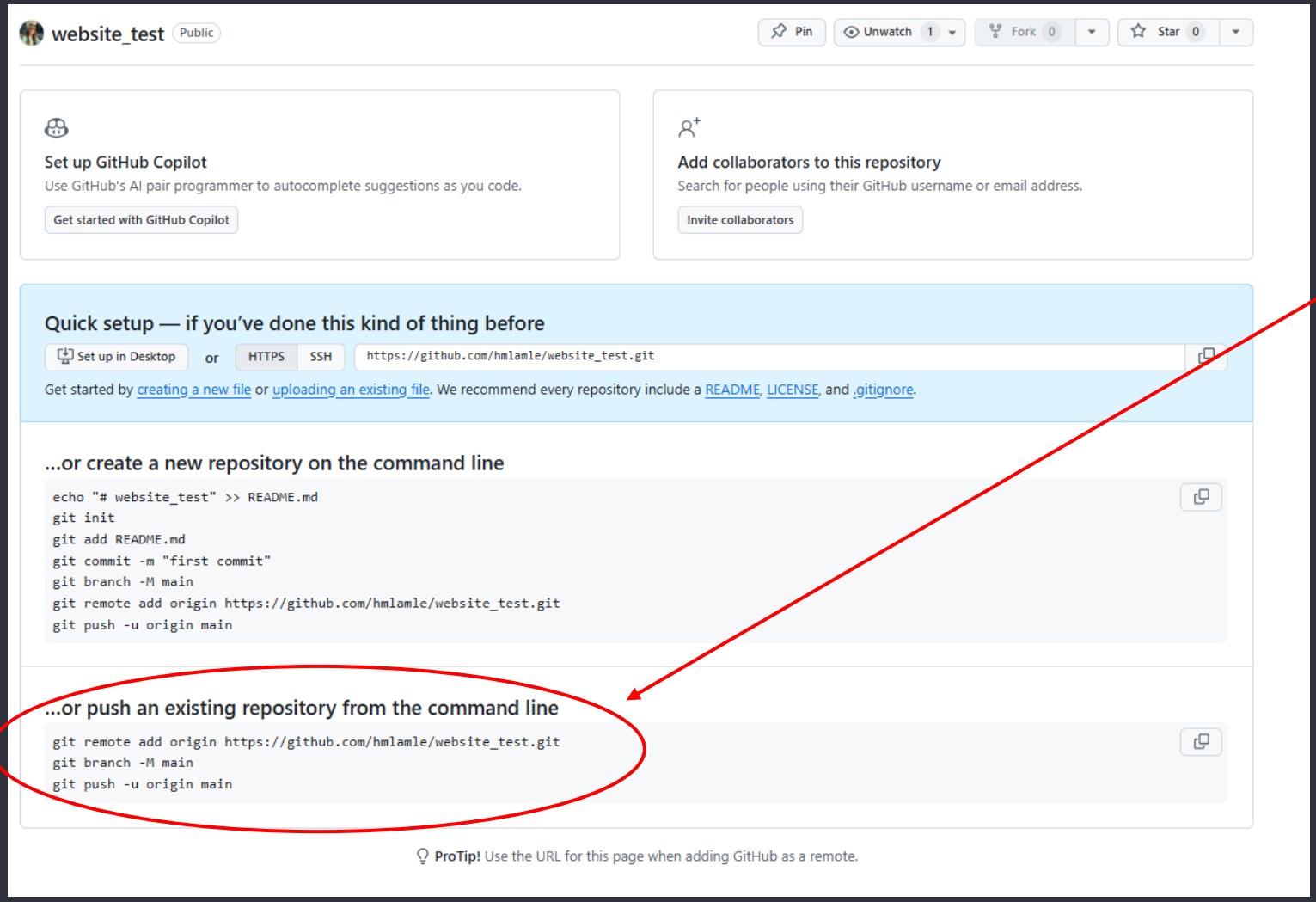
Name of repository will be website URL. Example:

Repository name: hmlamle.github.io
Website name: https://hmlamle.github.io/

Repository name: My_Crazy_Website
Website name: https://hmlamle.github.io/my_crazy_website/

Leave everything else here blank.

Set_Up.qmd

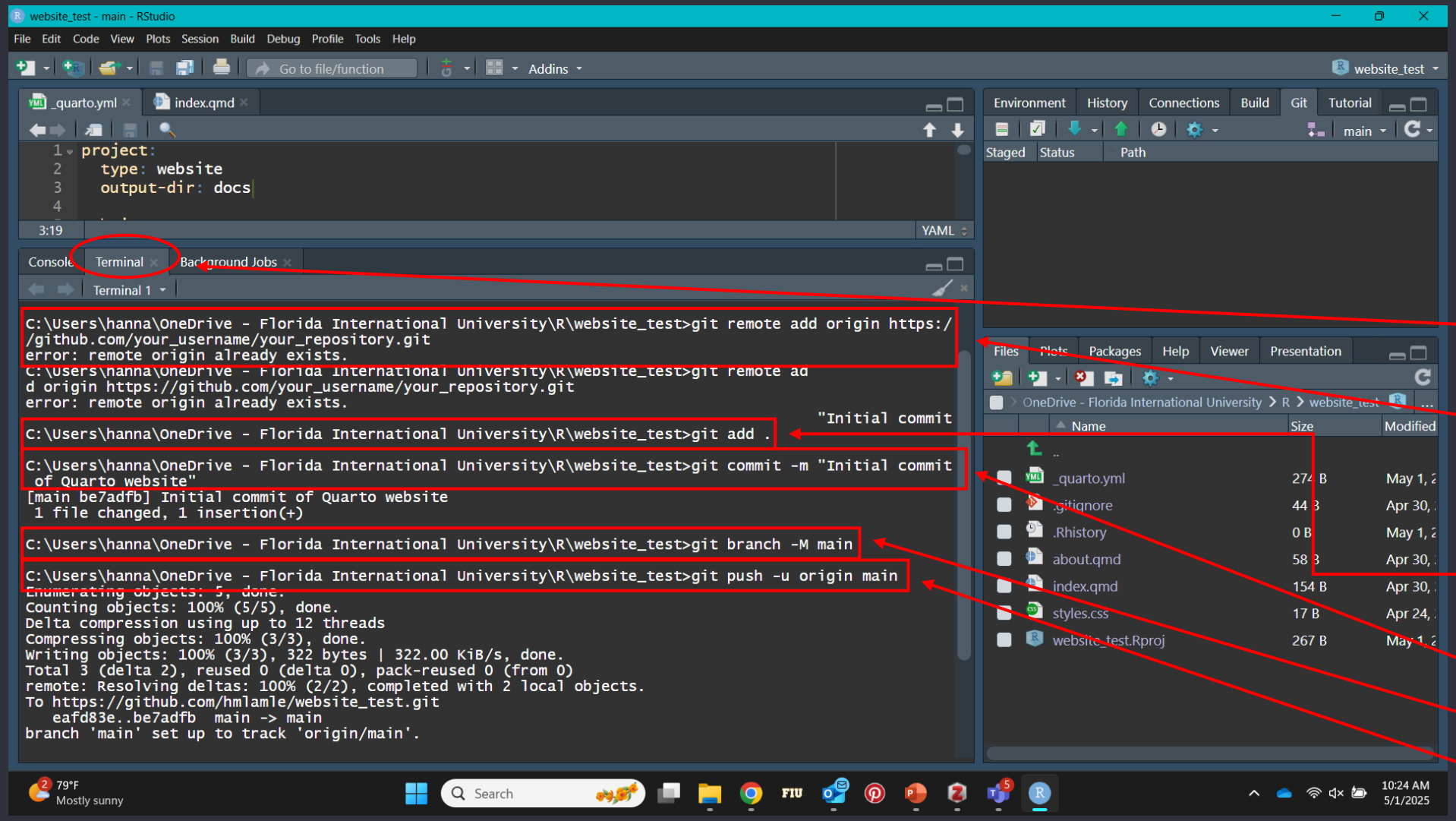


Since we already have an existing Git repo (we created one when we created the website in Rstudio) we just need to push it to our new GitHub repo…

Copy just the first line of code:

"git remote add origin https://github.com/your_username/your_repository.git"

Set_Up.qmd

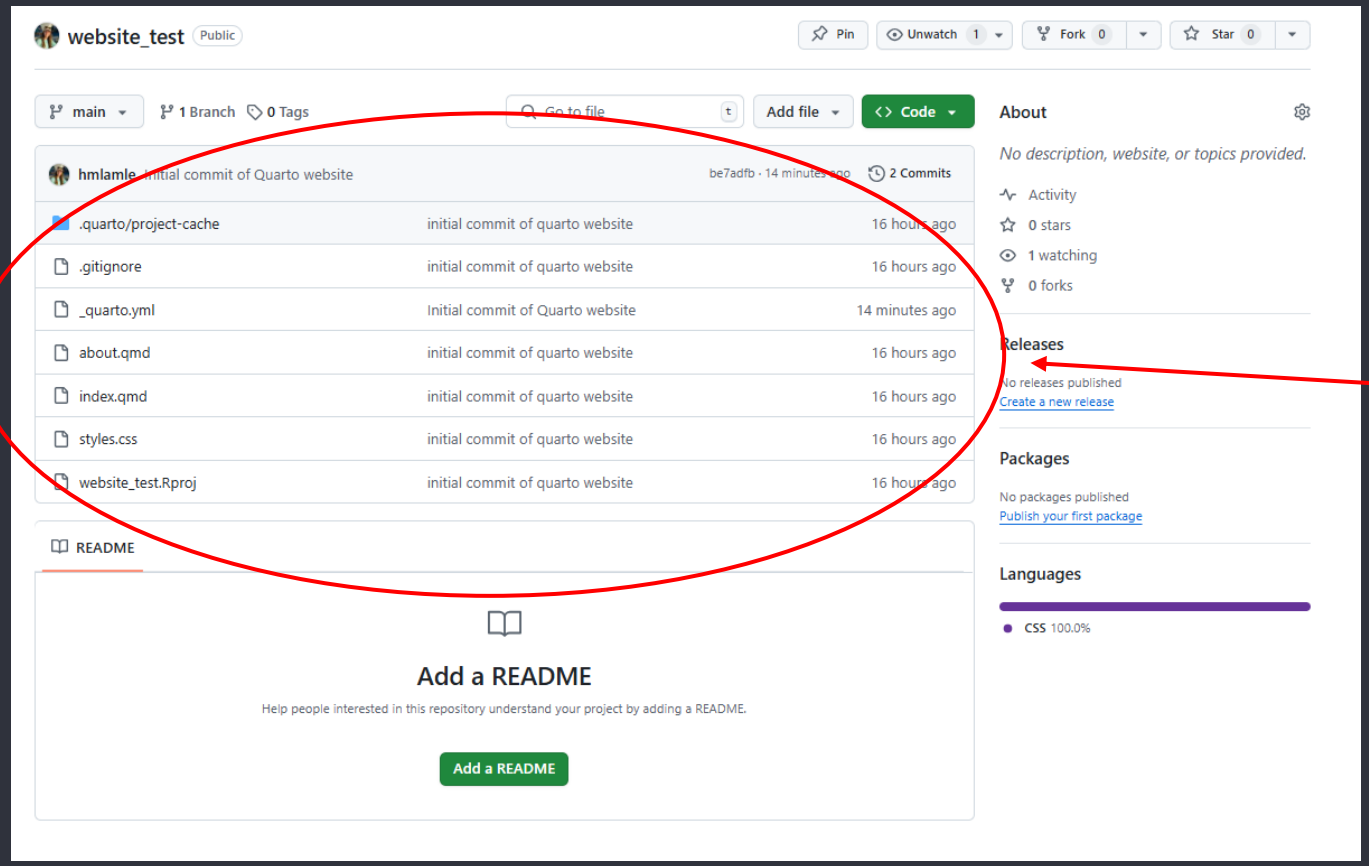


Back to Rstudio… finalizing the communication from Rstudio > Git > Github

1. Make sure you are in the terminal
2. Paste the code from GitHub. You should NOT see the error on my screen.
3. Type this code and enter. This is staging all our files.
4. Type this code and enter.
5. Type this code and enter.
6. Type this code and enter.

FCE_Science_Workshop_3.qmd

Back to GitHub AGAIN (last time I promise) to ensure communication...

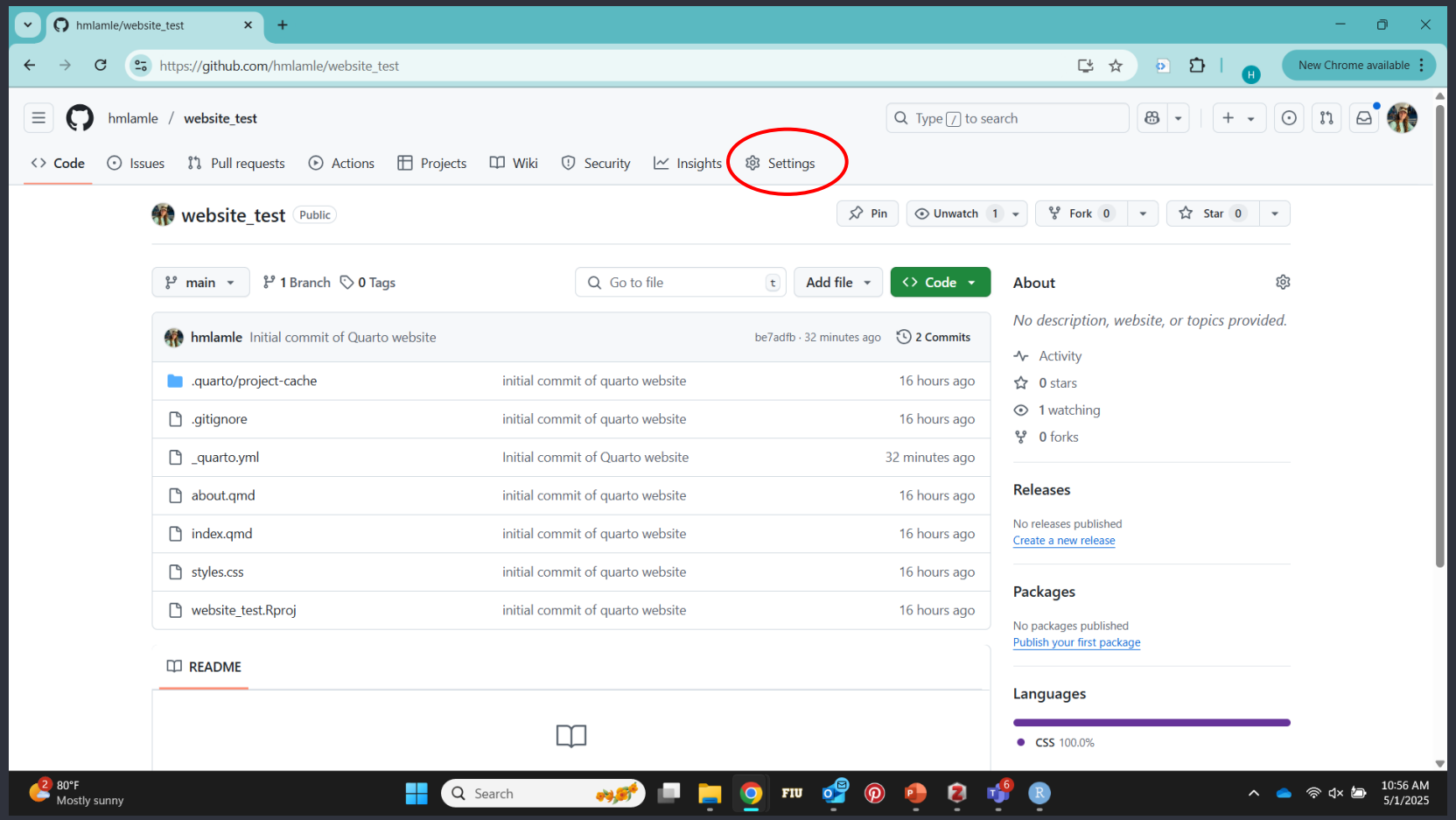


Should see all of our files now and time should be 1 minute ago (give or take)

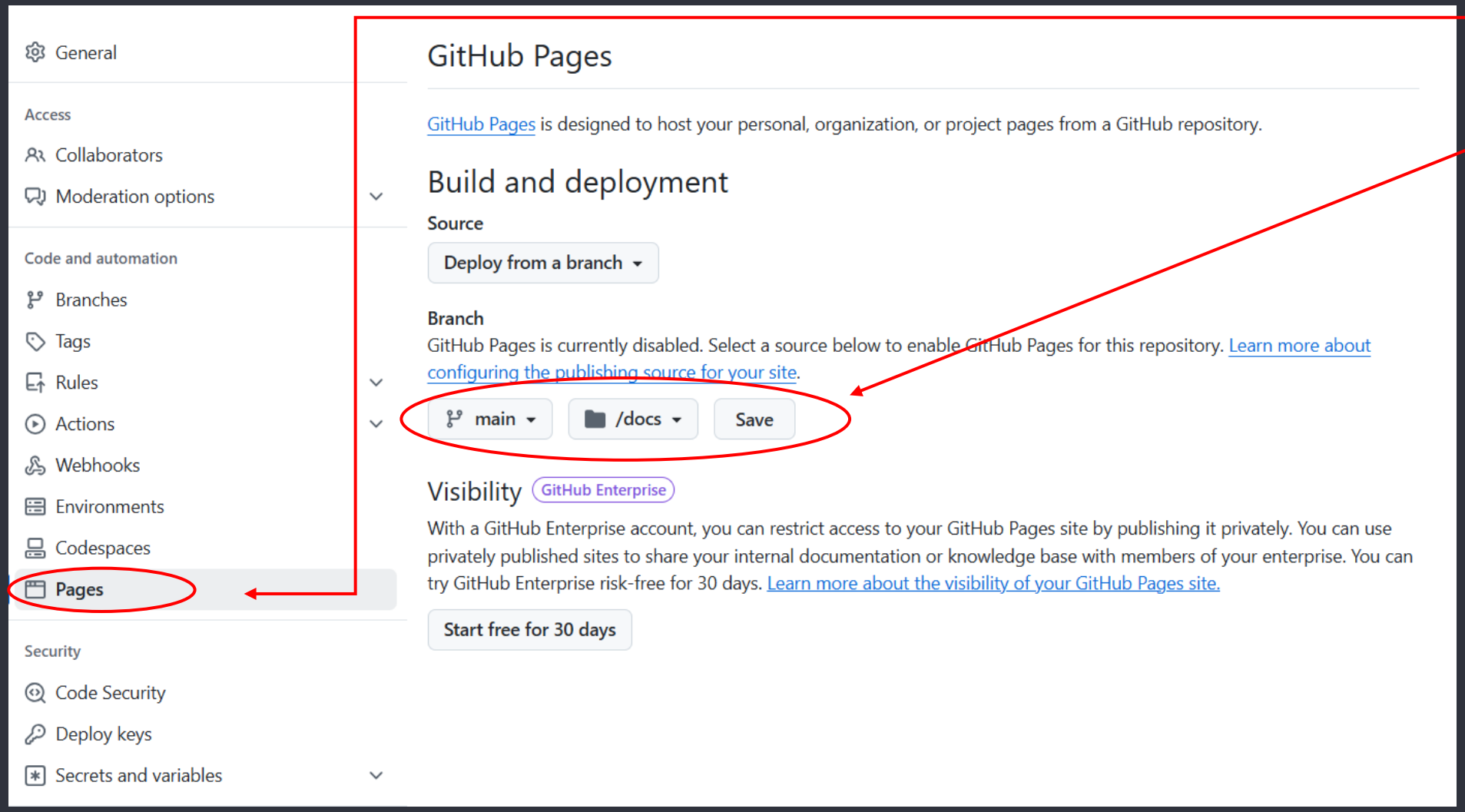
PAUSE... Quick review of what we just did:

1. **We downloaded Quarto (tool that renders documents from code and text)**
2. **We used Quarto to create a website for us in Rstudio and at the same time create a Git repo**
3. **We made some light edits to how the website renders to make it easier to host on GitHub (created docs folder)**
4. **We created our github repository to host our website**
5. **We connected our local website (on Rstudio) to Github (our online hosting service)**

And launch website!!



Set_Up.qmd



1. Go down to the pages tab

2. Click "none" and select "main"
3. Click /(root) and select /docs
4. save!

Did our website launch?? NO... why?

We haven't rendered it locally yet to create the docs folder!

Lets go back to Rstudio, customize our website and render it!

Quarto Website Files 101:

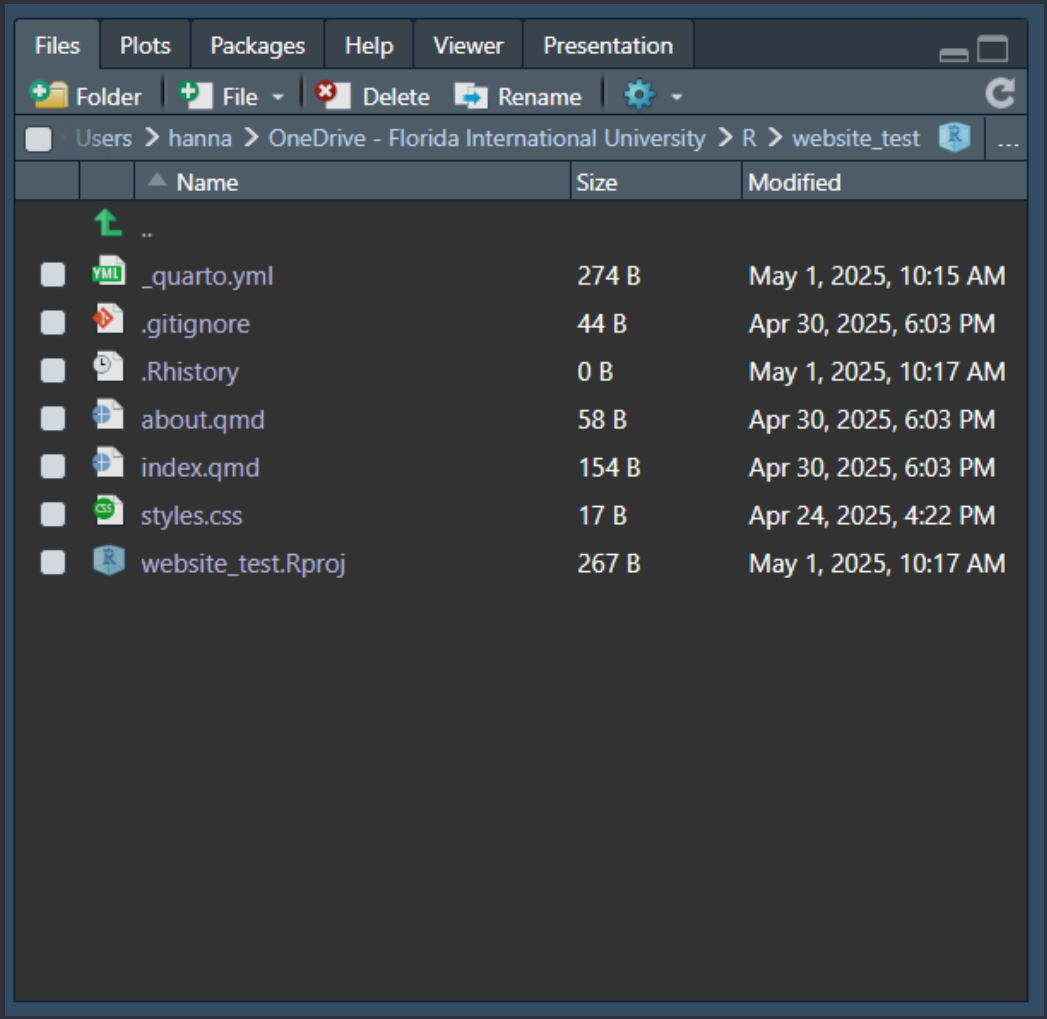
_quarto.yml - "settings" file for your website. Do not change name!

about.qmd - This is one page of our website! All pages will be Quarto markdown files.

index.qmd - Home/landing page of website.

styles.css - "cascading style sheet," custom formatting goes here.

.gitignore & .Rhistory – ignore, not needed for our purposes



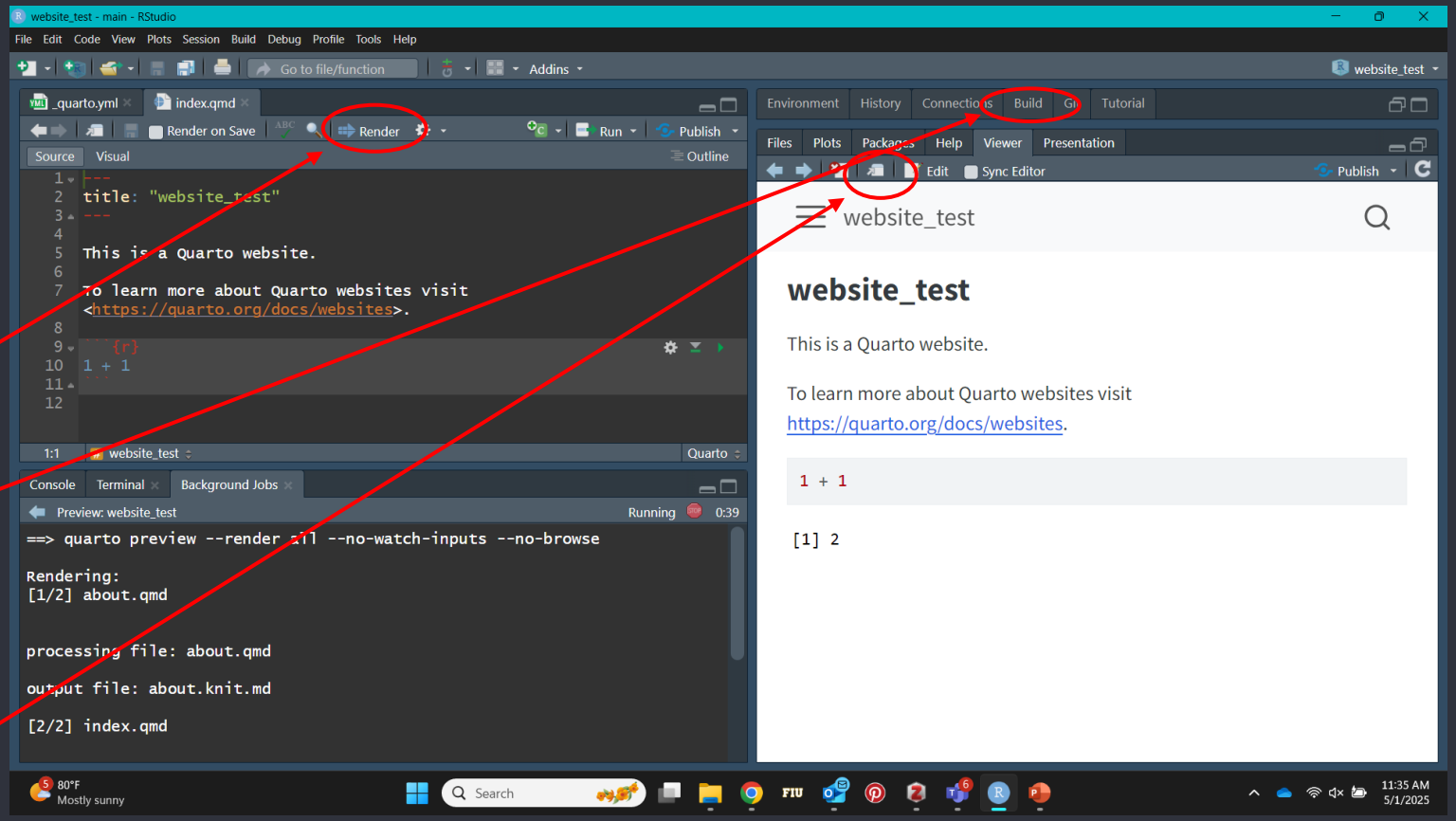
How do these files work together?

- Let's see by rendering our website:

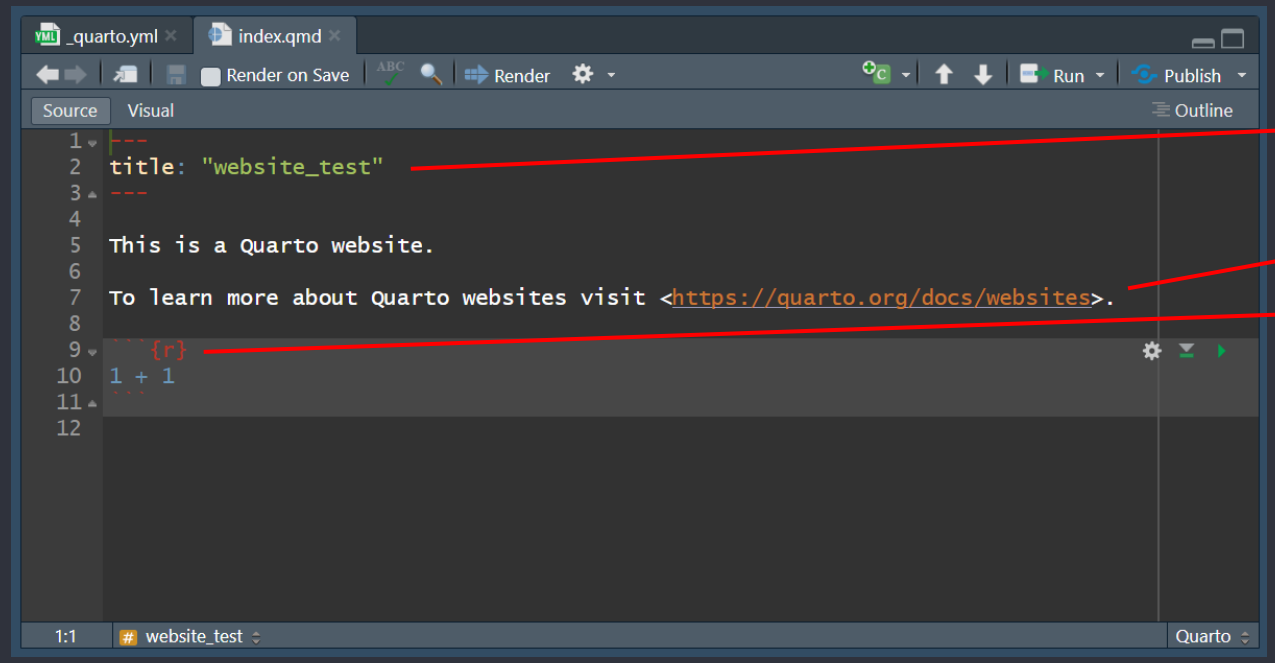
On a qmd file, this option just renders the page

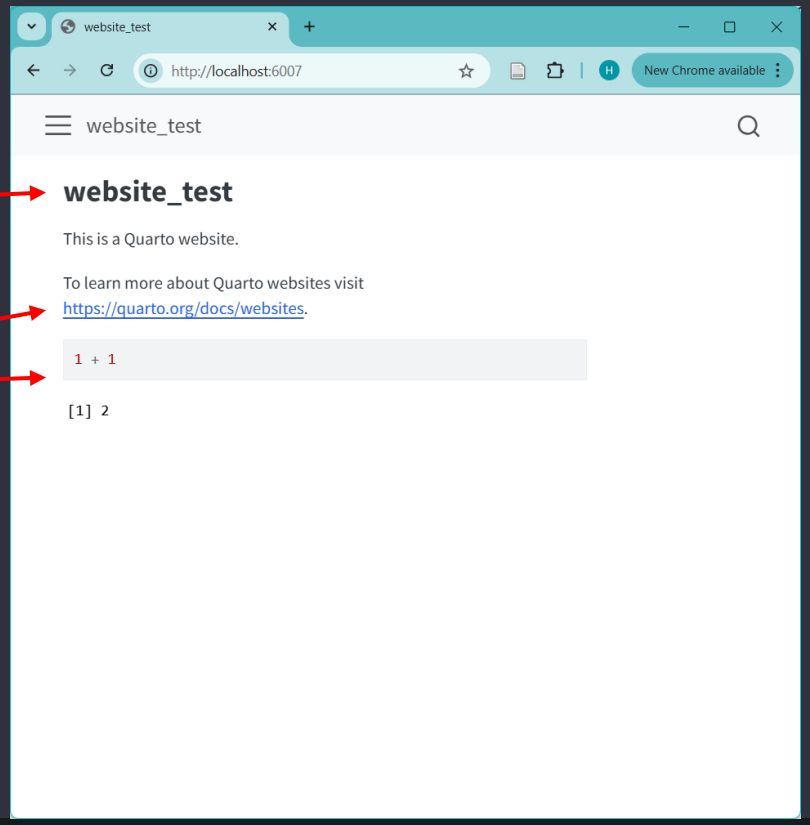
On "build" tab, will render entire website and place rendered html files in "docs" folder. DO THIS OPTION!

Your website will pop up in the viewer pane. Click this button and it will launch in your web browser instead (better viewing!)



Take a few minutes and click around on your website, and compare it to your files in Rstudio... this is the best way to learn the ins and outs of your website.

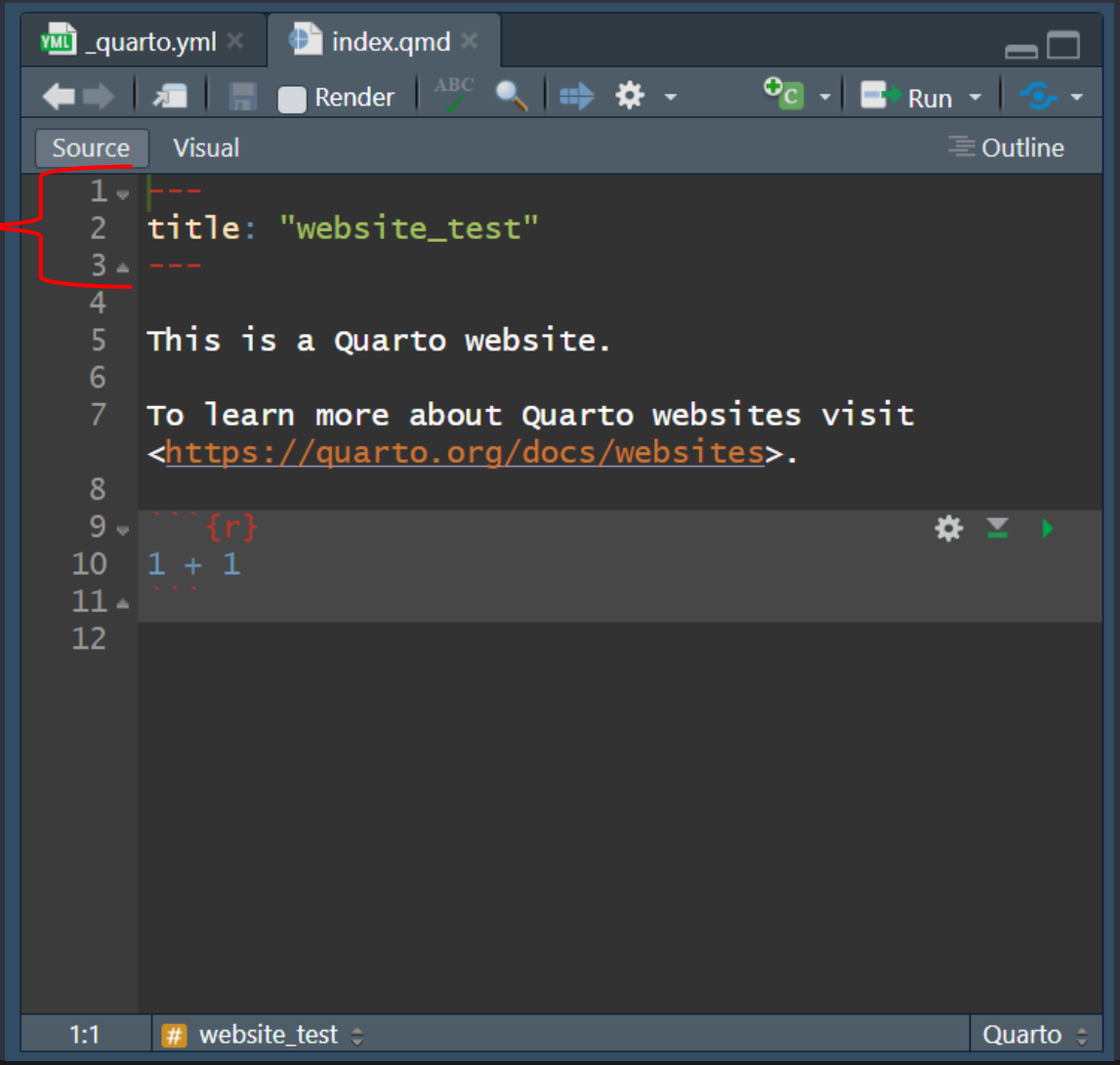




Page-level styling options…

Front matter – yaml block in qmd (same language as our settings file .yml).

- Why have it? Allows us to make page-level design specifications. Putting design in the .yml file will apply to the whole website
- Options for code to put here is quite large… not all are applicable for us with a website and may be better suited for publishing papers.
- See options here: https://quarto.org/docs/reference/formats/html.html



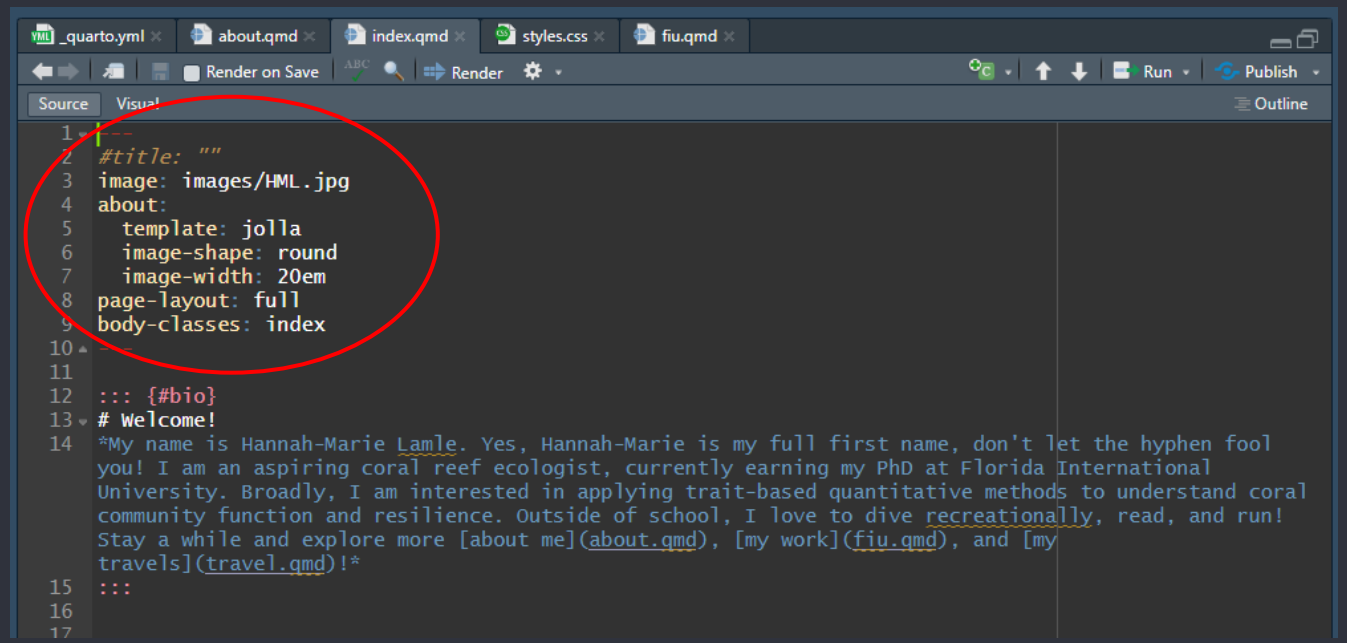
Page-level styling options… “About” page templates

- Quarto has a few pre-made templates that can be used to save time for designing.
- Don’t have to be used for your “about” page. I used mine in the index/landing page.

Options:

- jolla
- trestles
- solana
- marquee
- Broadside

https://quarto.org/docs/websites/website-about.html



Page-level styling options…

Webpage text – can do a lot here…

- Different headings (can go up to 5 subheadings, #####)

italics

bold

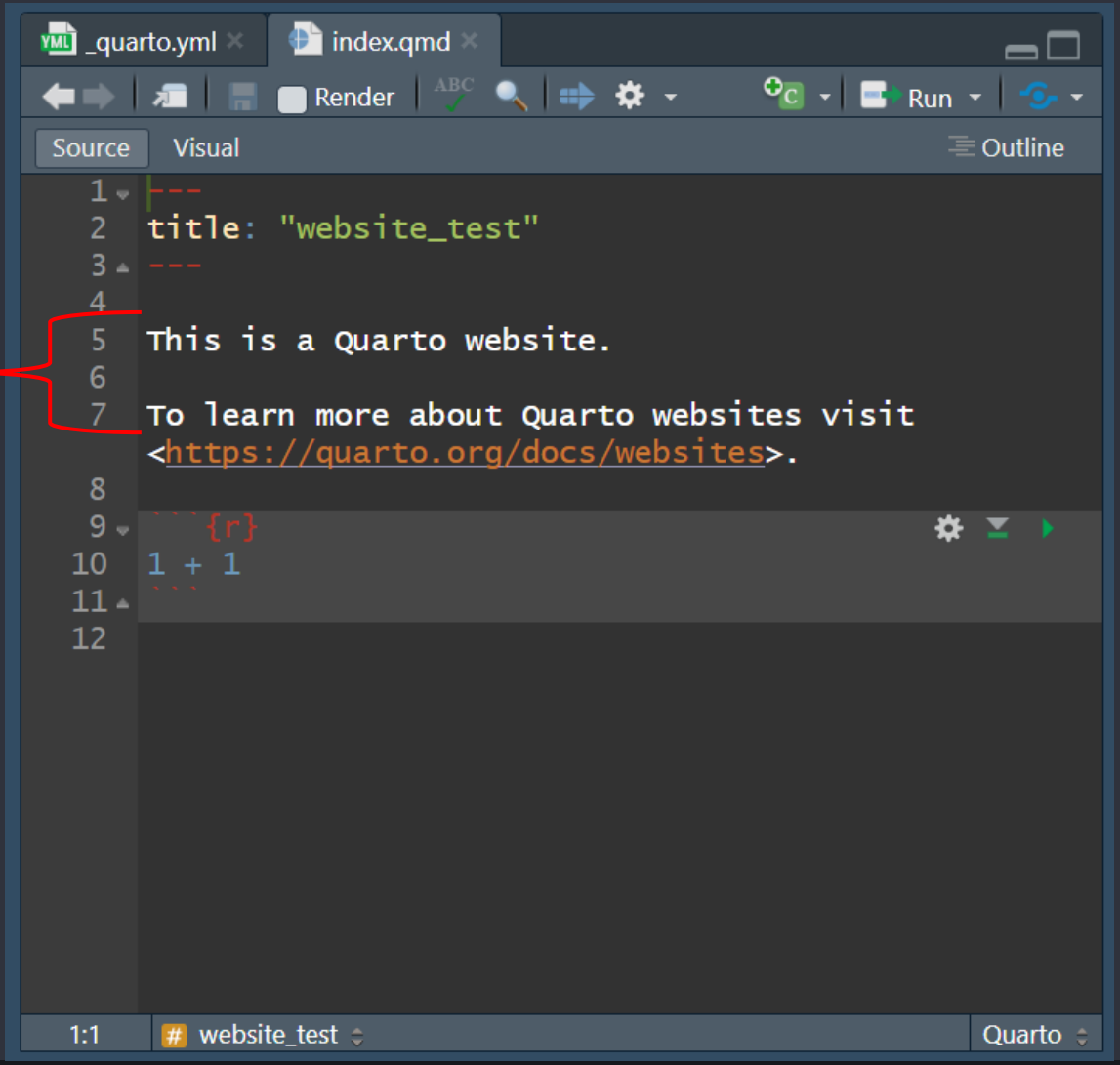
bold italics

<website link>

[text with link](https://quarto.org)

![Caption](image.png)

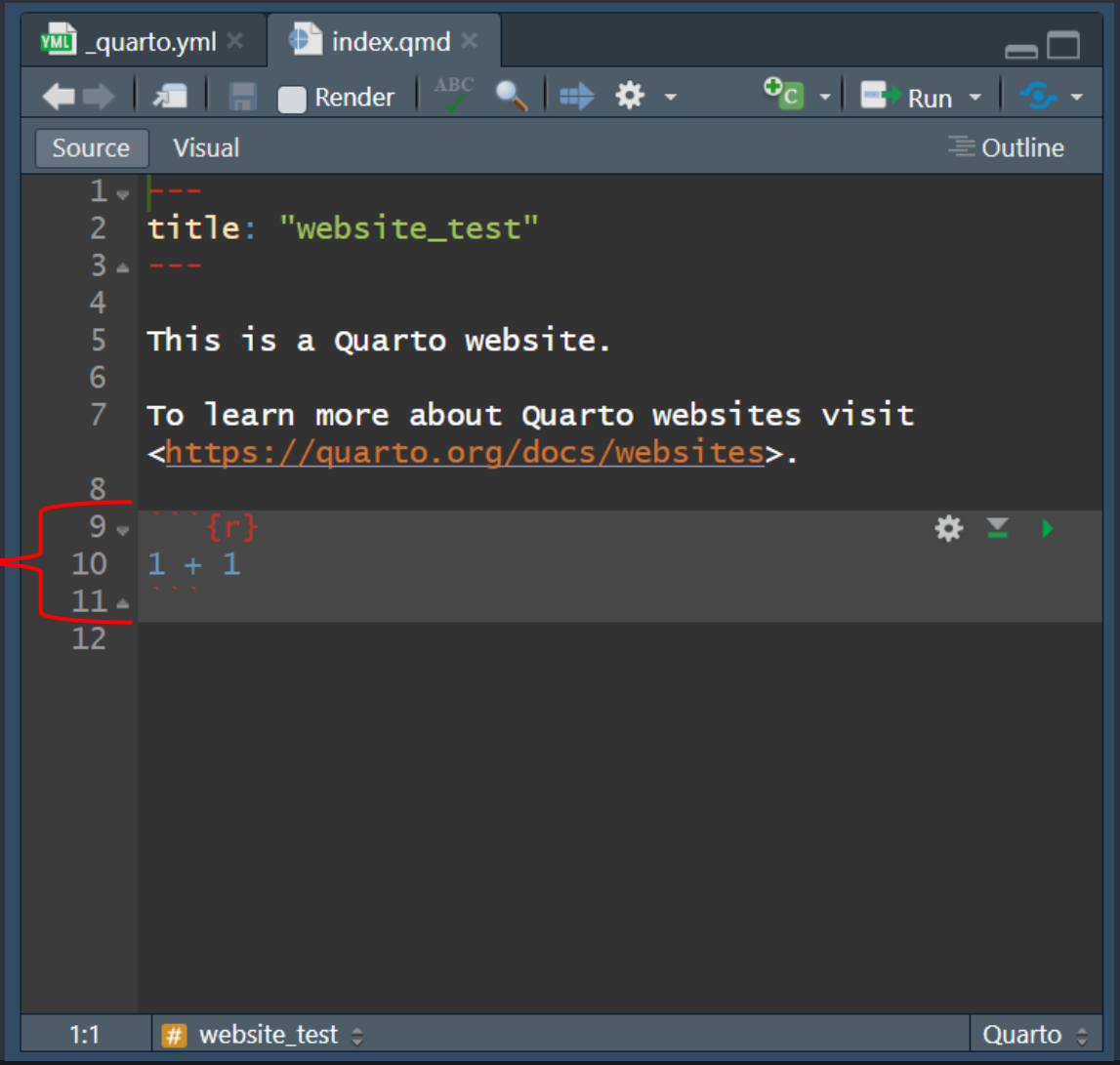
See more here: https://quarto.org/docs/authoring/markdown-basics.html



Page-level styling options…

Code chunk – can show off your coding skills with the raw code and the output, similar to markdown format.

- Code chunk does NOT have to be in R!!
- I use a lot of html code chunks in my lab's website to get custom formatting for my websites.
  - https://seascapes-fiu.github.io



Page-level styling options…

Code chunk – Does NOT have to be in R!!

- Can use divs and spans (html formatting language) to create "chunks" of your website that have different formatting.
  - The formatting for these chunks are controlled in your css sheet
- I use a lot of html code chunks in my lab's website to get advanced formatting for my websites.
  - https://seascapes-fiu.github.io

styles.css | about.qmd

Render on Save | Render | Run | Publish

Source | Visual | Outline

```
---
title: "About Me"
format: html
css: styles.css
page-layout: full
body-classes: about
---


### Upbringing
![](images/aboutme1.jpg){.img-right}


::: {.larger-text}
I was born and raised in Northern Indiana, a product of [Land of Lakes County](https://www.travelindiana.com/kosciusko-county-the-land-of-lakes/). I only half jokingly say that lake water makes up my blood. Growing up, my family and I would spend every possible moment on our boat. I have fond memories of enjoying the water: from leisurely boat rides, dropping anchor and getting lost in a book, to water skiing, wakeboarding, and surfing. I even got my PADI open water certification in the lake! Each summer before high school (before sports practices started to dominate my schedule) I would also commandeer my families paddle boat to catch turtles, jar seaweed to ID on the family computer, and look for bluegill nests along the channel edge. Being so connected to the water from an early age undoubtedly shaped my career path in the natural sciences.
:::


### Hobbies
::: {.larger-text}
In addition to loving the outdoors, I grew up playing many sports in school, and sports are still a large part of my life today. I grew up playing tennis, and then played basketball, volleyball, and ran track in high school, so being active is a large part of who I am. I also was on my collegiate wakeboarding club team, which was such as cool experience and how I met some of my best friends at university. I'm a passionate fan of men's and women's basketball, always rooting on my Hoosiers, and love March Madness. I don't play a lot of basketball anymore, but do spend a lot of my free time playing tennis recreationally. It's a lot easier to do so here in South Florida, where there's never any snow! I also have recently been getting into running as a hobby, and completed my first half marathon in February of 2025. Looking forward to more races to come!
```

28:19 | Scuba Diving | Quarto

Styles.css file:

Notice that the styles.css file is automatically applied in the yml, but sheet itself is empty.

- **Different language!**
- **More recently, been seeing people's websites using .scss files, and Quarto website encourages use (https://quarto.org/docs/output-formats/html-themes.html#theme-options)**
- **Work in progress for me to shift websites over to .scss file.**
  - https://github.com/twbs/bootstrap/blob/main/scss/_variables.scss

styles.css | about.qmd

```
/* ---------------------------- General Formatting ------------------------------*/

.img-right {
  float: right;
  margin: 20px auto;
  width: 40%;
}

.img-left {
  float: left;
  margin: auto 20px;
  width: 40%;
}

.larger-text {
  font-size: clamp(1rem, 1.2vw, 2.5rem);
}

html, body.about {
  height: auto; /* Allows the background to expand beyond viewport */
```

107:1 CSS

Design.qmd

about.qmd

styles.css | about.qmd

Render on Save | Render | Run | Publish

Source | Visual | Outline

```
---
title: "About Me"
format: html
css: styles.css
page-layout: full
body-classes: about
---


### Upbringing
![](images/aboutme1.jpg){.img-right}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```

28:19 | Scuba Diving | Quarto

styles.css

styles.css | about.qmd

```
  font-size: 2em;  /* Adjust the font size of the bio text */
  max-width: 600px;  /* Set a max width for readability */
  padding: 0 20px;  /* Add padding to the sides of the bio text */
}


/* ---------------------------- General Formatting ----------------


.img-right {
  float: right;
  margin: 20px auto;
  width: 40%;
}

.img-left {
  float: left;
  margin: auto 20px;
  width: 40%;
}

.larger-text {
  font-size: clamp(1rem, 1.2vw, 2.5rem);
}


body.about {
  background-image: url("images/about_background.png");
  background-repeat: no-repeat;
  background-size: cover;
  background-position: center top;
  width: 100%;
  min-height: 100vh;
  margin: 0;
  padding: 0;
  color: #1d3a69;
}



/* ---------------------------- IU Research ----------------------
/* not using this page at the moment
```

123:1 | CSS

about.html

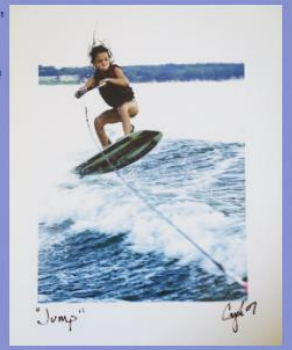
Files | Plots | Packages | Help | Viewer | Presentation

Edit | Sync Editor | Publish

Hannah Marine

About Me

Upbringing



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FCE_Science_Workshop_3.qmd

Quick example (you try too!):

website_test - main - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

_quarto.yml | index.qmd | styles.css

```
/* css styles */

.navbar {
  background-color: #b9ff66;
}

.body {
  background-color: tan;
}
```

5:2 CSS

Console | Terminal | Background Jobs

R 4.4.1 · C:/Users/hanna/OneDrive - Florida International University/R/website_test/

```
  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

>
```

Environment | History | Connections | Build | Git | Tutorial

Files | Plots | Packages | Help | Viewer | Presentation

Edit | Sync Editor | Publish

website_test

website_test

This is a Quarto website.

To learn more about Quarto websites visit https://quarto.org/docs/websites.

```
1 + 1
```

```
[1] 2
```

High UV Now | Search | 1:12 PM 5/1/2025

YAML styling options…

- We already changed this once to include the output directory to be “docs” (compatible with GitHub pages)
- Global (whole website) design alterations
- There are many more options to make websites your own:
  - Designing the navbar
  - Including pages
  - Adding a footer
  - Changing the title and logo
  - Links to your socials, GitHub, etc
  - Even more, found here: https://quarto.org/docs/reference/projects/websites.html

_quarto.yml | index.qmd | styles.css

```yaml
project:
  type: website
  output-dir: docs

website:
  title: "website_test"
  navbar:
    left:
      - href: index.qmd
        text: Home
      - about.qmd
      - href: FCE_dashv1.html
        text: Data Dashboard

format:
  html:
    theme:
      - cosmo
      - brand
    css: styles.css
    toc: true
```

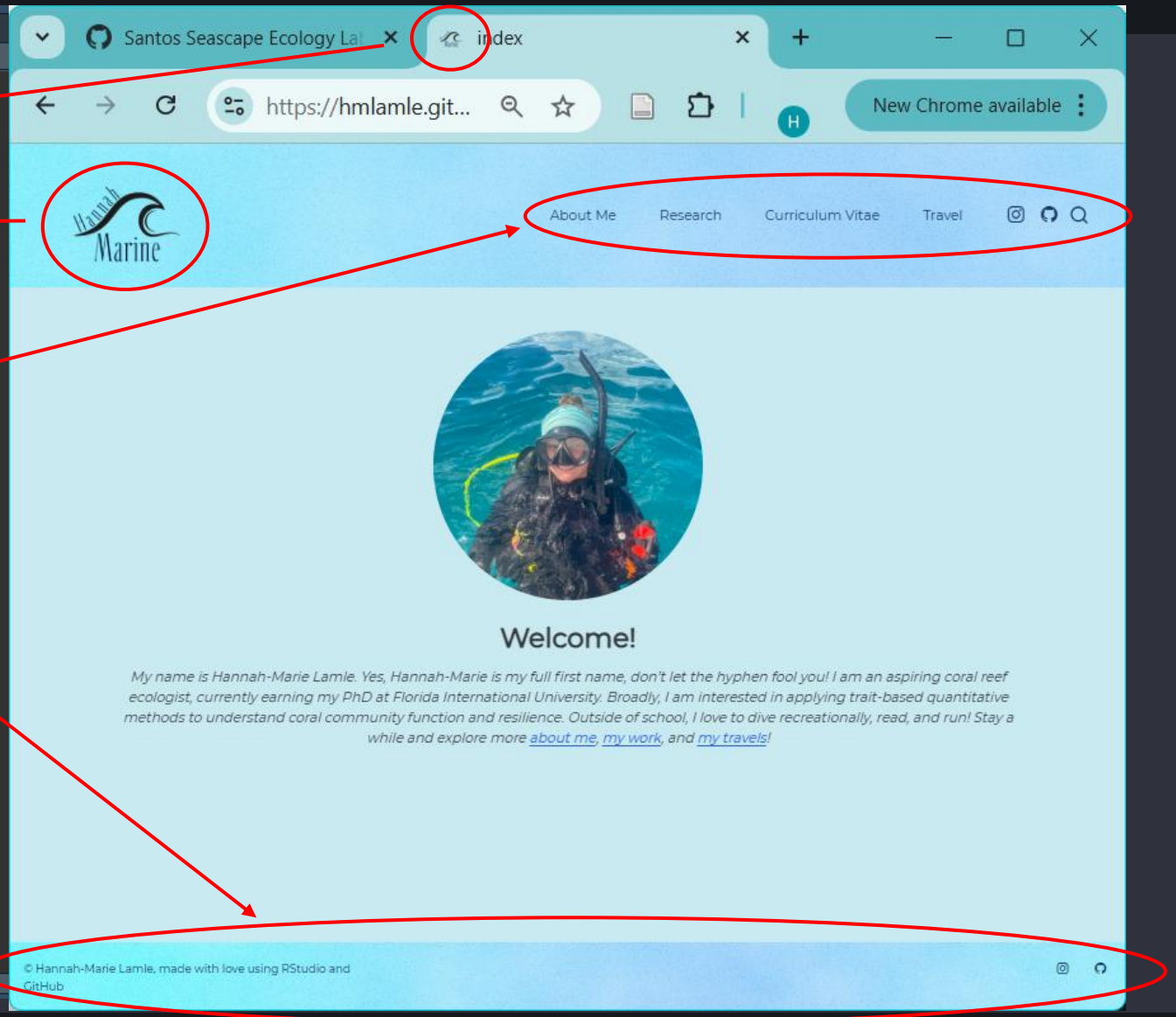
22:1 YAML

Design.qmd

_quarto.yml | styles.css | fiu.qmd

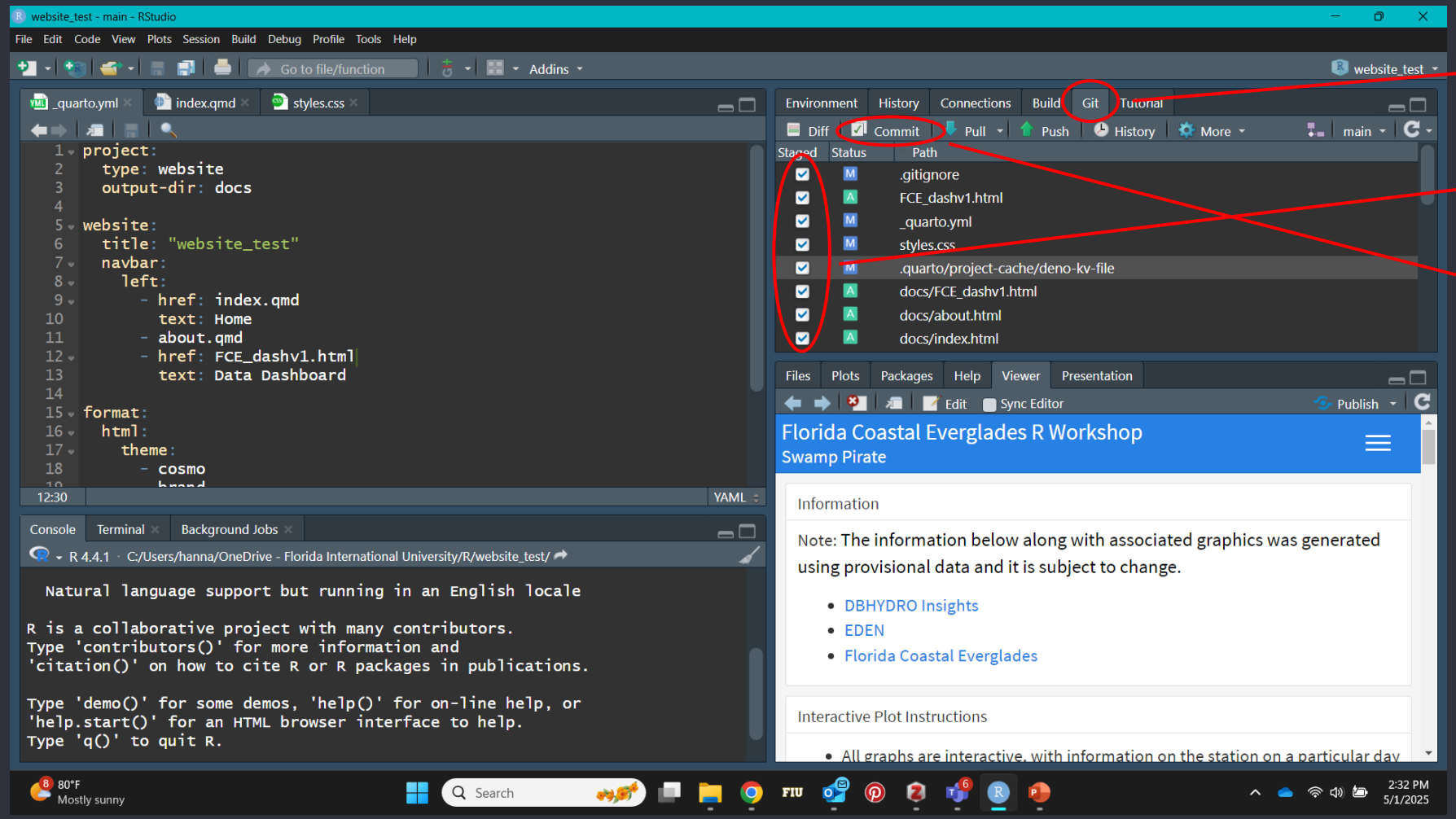
```yaml
project:
  type: website
  output-dir: docs

website:
  favicon: 'images/logo.png'
  navbar:
    logo: 'images/logo.png'
    right:
      - about.qmd
      - text: "Research"
        href: fiu.qmd
      - CV.qmd
      - text: "Travel"
        href: travel.qmd
    tools:
      - icon: instagram
        href: https://www.instagram.com/hannah_mari.n.e/
      - icon: bluesky
        href: https://bsky.app/profile/hannah-marine.bsky.social
      - icon: github
        href: https://github.com/hmlamle
    toggle-position: right
    tools-collapse: true

  page-footer:
    left: © Hannah-Marie Lamle, made with love using RStudio and GitHub
    right:
      - icon: instagram
        href: https://www.instagram.com/hannah_mari.n.e/
      - icon: bluesky
        href: https://bsky.app/profile/hannah-marine.bsky.social
      - icon: github
        href: https://github.com/hmlamle

format:
  html:
    theme: cosmo
    css: styles.css
    toc: false
```

11:24



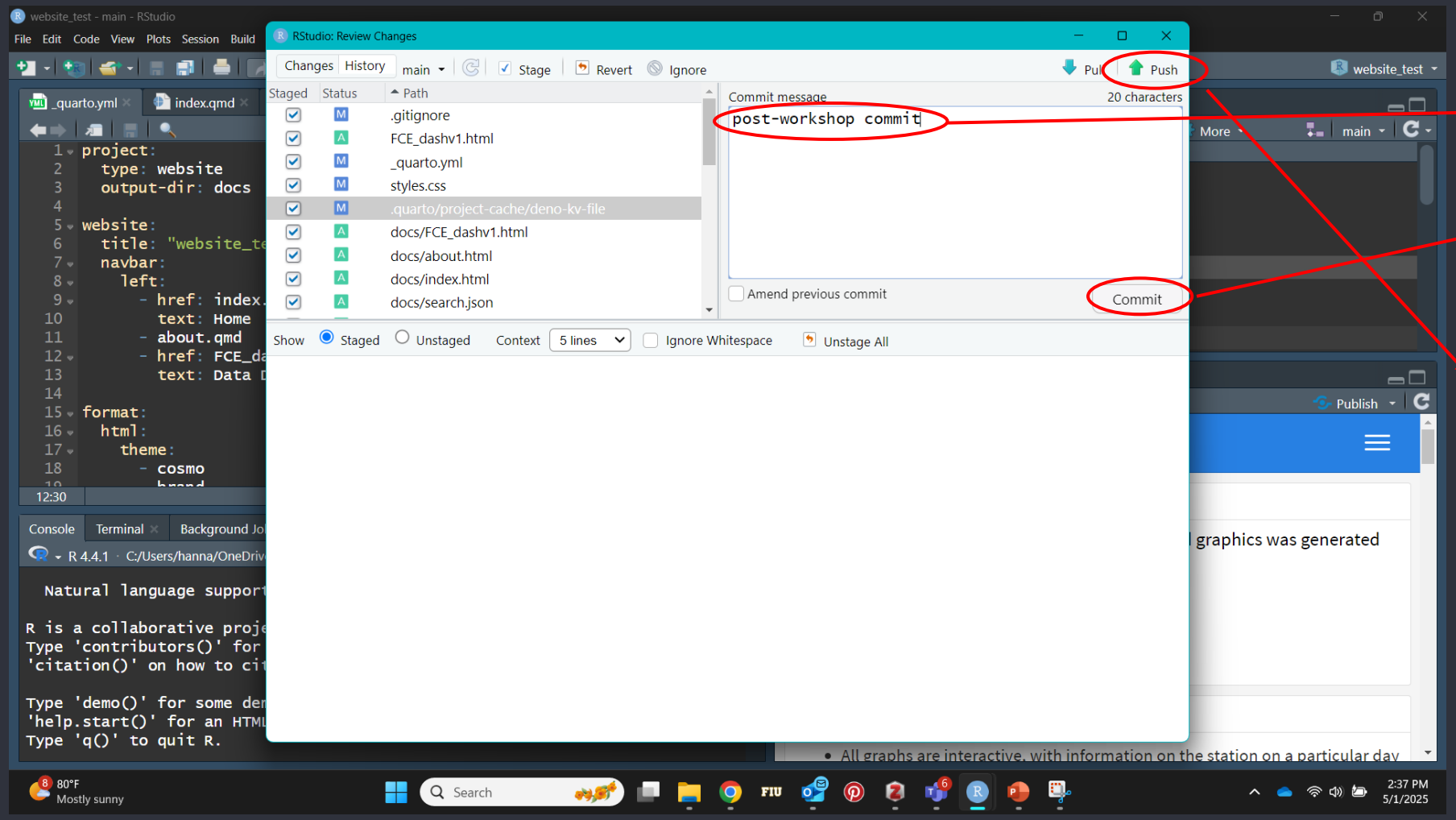
FCE_Science_Workshop_3.qmd

Done? Save, commit, push changes!



1. Go to Git tab
2. Check all files
3. Commit changes

Done? Save, commit, push changes!



1. Write commit message (the shorter the better)

2. Commit

3. Push changes

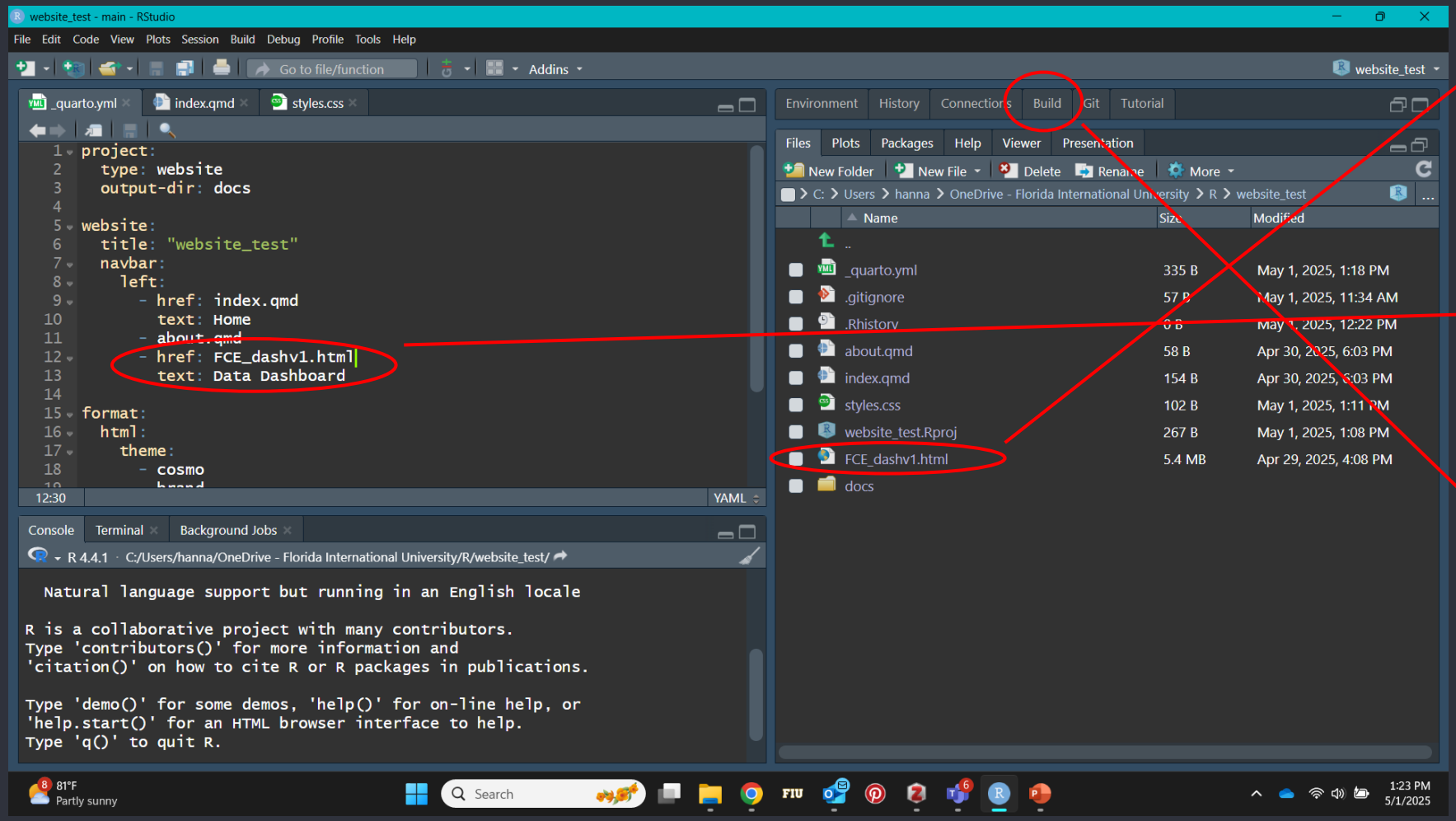
4. (not shown) go back to GitHub and check your deployments, should not fail now and you will have a working website!

Resources.qmd

```
# Thank you for listening! Happy website making ☺

# Resources:
        # website:                                  Github code:
    https://hmlamle.github.io/                https://github.com/hmlamle/hmlamle.github.io
    https://seascapes-fiu.github.io/          https://github.com/Seascapes-FIU/seascapes-fiu.github.io
    https://mackwhite.github.io/              https://github.com/mackwhite/mackwhite.github.io
    https://samanthacsik.github.io/           https://github.com/samanthacsik/samanthacsik.github.io
    https://eric.walkingshaw.net/             https://github.com/walkie/walkie.github.io

# How to style your quarto docs without knowing HTML
# Slide deck for Advanced Quarto formatting

# Questions?
```

FCE_Science_Workshop_5.qmd

BONUS: Adding your r dashboard to your website?



1. Copy your dashboard html file into your working directory
2. Add it to the navbar in the yml file
3. Build website..